LOCAL COHOMOLOGY WITH RESPECT TO TWO IDEALS AND THEIR SUM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of local cohomology through the extended Cech complex on generators of the ideal. Invoked inside the counted proof (source lines 1678-1683). Turns the composition into a statement about tensor products of Cech complexes.

Source reference. Original reference derived-lemma-grothendieck-spectral-sequence: sources/stacks-dualizing/pinned-original-source.tex, source lines 1664–1664. Navigation pointer only: the counted text refers to the Stacks item derived-lemma-grothendieck-spectral-sequence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-local-cohomology-adjoint: sources/stacks-dualizing/pinned-original-source.tex, source lines 1466–1466. Navigation pointer only: the counted text refers to the Stacks item lemma-local-cohomology-adjoint, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let A be a ring and let $I, J \subset A$ be finitely generated ideals. Set $Z = V(I)$ and $Y = V(J)$. Then $Z \cap Y = V(I + J)$ and $R\Gamma_Y \circ R\Gamma_Z = R\Gamma_{Y \cap Z}$ as functors $D(A) \to D_{(I+J)^\infty\text{-}torsion}(A)$. For $K \in D^+(A)$ there is a spectral sequence*

$$E_2^{p,q} = H_Y^p(H_Z^q(K)) \Rightarrow H_{Y \cap Z}^{p+q}(K)$$

as in Derived Categories, Lemma derived-lemma-grothendieck-spectral-sequence.

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Proof. There is a bit of abuse of notation in the lemma as strictly speaking we cannot compose $R\Gamma_Y$ and $R\Gamma_Z$. The meaning of the statement is simply that we are composing $R\Gamma_Z$ with the inclusion $D_{I^\infty\text{-torsion}}(A) \to D(A)$ and then with $R\Gamma_Y$. Then the equality $R\Gamma_Y \circ R\Gamma_Z = R\Gamma_{Y \cap Z}$ follows from the fact that

$$D_{I^\infty\text{-torsion}}(A) \to D(A) \xrightarrow{R\Gamma_Y} D_{(I+J)^\infty\text{-torsion}}(A)$$

is right adjoint to the inclusion $D_{(I+J)^\infty\text{-torsion}}(A) \to D_{I^\infty\text{-torsion}}(A)$. Alternatively one can prove the formula using Lemma lemma-local-cohomology-adjoint and the fact that the tensor product of extended Čech complexes on $f_1, \ldots, f_r$ and $g_1, \ldots, g_m$ is the extended Čech complex on $f_1, \ldots, f_n.g_1, \ldots, g_m$. The final assertion follows from this and the cited lemma. $\square$